

I n t r o d u c t i o n

Hash Tables

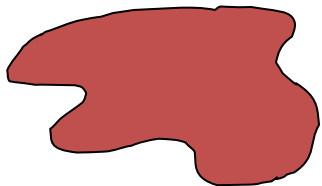
Data Structures and Algorithms

Hash Tables

Constant time accesses!

A **hash table** is an array of some fixed size, usually a prime number.

General idea:



key space (e.g., integers, strings)

hash function:

$h(K)$



hash table

0



TableSize - 1

Example

key space = integers

TableSize = 10

$$h(K) = K \bmod 10$$

Insert: 7, 18, 41, 94

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

Another Example

key space = integers

TableSize = 6

$h(K) = K \bmod 6$

Insert: 7, 18, 41, 34

0	
1	
2	
3	
4	
5	

Direct Addressing

- Need a data structure for C
- There are 2^{32} different IP(v4) addresses
- Convert IP to 32-bit integer

Direct Addressing

- There are 2^{32} different IP(v4)
- Convert IP to 32-bit integer
- Create an integer array A of size 2^{32}

Encoding IPs

- Encode IPs with small numbers
- i.e. numbers from 0 to 999
- Different codes for currently active IPs

Hash Function

Definition

For any set of objects S and any integer $m > 0$, a function $h : S \rightarrow \{0, 1, \dots, m - 1\}$ is called a **hash function**.

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Definition

m is called the **cardinality** of hash function h .

Desirable Properties

- h should be fast to compute
- Different values for different objects
- Direct addressing with $O(m)$ memory
- Want small cardinality m
- Impossible to have all different values if number of objects $|S|$ is more than m

Popular Hash Function

Division method: Choose a number m larger than the number n of keys in K . (The number m is usually chosen to be a prime number or a number without small divisors, since this frequently minimizes the number of collisions.) The hash function H is defined by

$$H(k) = k \text{ (mod } m) \text{ or } H(k) = k \text{ (mod } m) + 1$$

Here $k \text{ (mod } m)$ denotes the remainder when k is divided by m . The second formula is used when we want the hash addresses to range from 1 to m rather than from 0 to $m - 1$.

Popular Hash Function

Midsquare method: The key k is squared. Then the hash function H is defined by

$$H(k) = l$$

where l is obtained by deleting digits from both ends of k^2 . We emphasize that the same positions of k^2 must be used for all of the keys.

Popular Hash Function

Folding method: The key k is partitioned into a number of parts, $k_1, \dots, k_r$, where each part, except possibly the last, has the same number of digits as the required address. Then the parts are added together, ignoring the last carry. That is,

$$H(k) = k_1 + k_2 + \dots + k_r$$

where the leading-digit carries, if any, are ignored. Sometimes, for extra “milling,” the even-numbered parts, $k_2, k_4, \dots$, are each reversed before the addition.

Question

Suppose a company with 68 employees assigns a 4-digit employee number to each employee which is used as the primary key in the company's employee file. Suppose L consists of 100 two-digit addresses: 00, 01, 02, ..., 99. Compute the locations to which the keys 3205, 7148, and 2345 are mapped.

(a) Division method. Choose a prime number m close to 99, such as $m = 97$.

Then

$$H(3205) = 4, H(7148) = 67, H(2345) = 17$$

That is, dividing 3205 by 97 gives a remainder of 4, dividing 7148 by 97 gives a remainder of 67, and dividing 2345 by 97 gives a remainder of 17. In the case that the memory addresses begin with 01 rather than 00, we choose that the function $H(k) = k(\bmod m) + 1$ to obtain:

$$H(3205) = 4 + 1 = 5, H(7148) = 67 + 1 = 68, H(2345) = 17 + 1 = 18$$

(b) Midsquare method. The following calculations are performed:

k :	3205	7148	2345
k^2 :	10 272 025	51 093 904	5 499 025
$H(k)$:	72	93	99

Observe that the fourth and fifth digits, counting from the right, are chosen for the hash address.

Question

Suppose a company with 68 employees assigns a 4-digit employee number to each employee which is used as the primary key in the company's employee file. Suppose L consists of 100 two-digit addresses: 00, 01, 02, ..., 99. Compute the locations to which the keys 3205, 7148, and 2345 are mapped.

(c) *Folding method.* Chopping the key k into two parts and adding yields the following hash addresses:

$$H(3205) = 32 + 05 = 37, H(7148) = 71 + 48 = 19, H(2345) = 23 + 45 = 68$$

Observe that the leading digit 1 in $H(7148)$ is ignored. Alternatively, one may want to reverse the second part before adding, thus producing the following hash addresses:

$$H(3205) = 32 + 50 = 82, H(7148) = 71 + 84 = 55, H(2345) = 23 + 54 = 77$$

Question

Consider a hash table of size $m=1000$ and a corresponding hash function $h(k)=\lfloor m(kA \bmod 1) \rfloor$ for $A=(\sqrt{5}-1)/2$. Compute the locations to which the keys 6161, 6262, 6363, 6464, and 6565 are mapped.

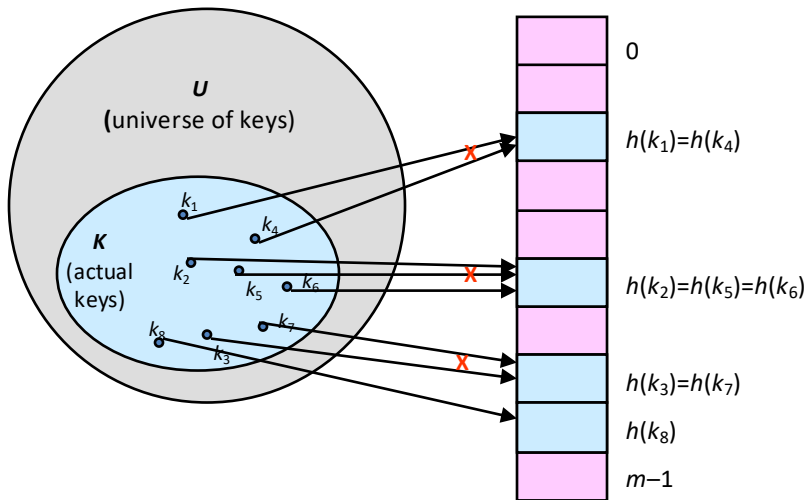
- $h(61) = \lfloor 1000(61 \cdot \frac{\sqrt{5}-1}{2} \bmod 1) \rfloor = 700.$
- $h(62) = \lfloor 1000(62 \cdot \frac{\sqrt{5}-1}{2} \bmod 1) \rfloor = 318.$
- $h(63) = \lfloor 1000(63 \cdot \frac{\sqrt{5}-1}{2} \bmod 1) \rfloor = 936.$
- $h(64) = \lfloor 1000(64 \cdot \frac{\sqrt{5}-1}{2} \bmod 1) \rfloor = 554.$
- $h(65) = \lfloor 1000(65 \cdot \frac{\sqrt{5}-1}{2} \bmod 1) \rfloor = 172.$

Collisions

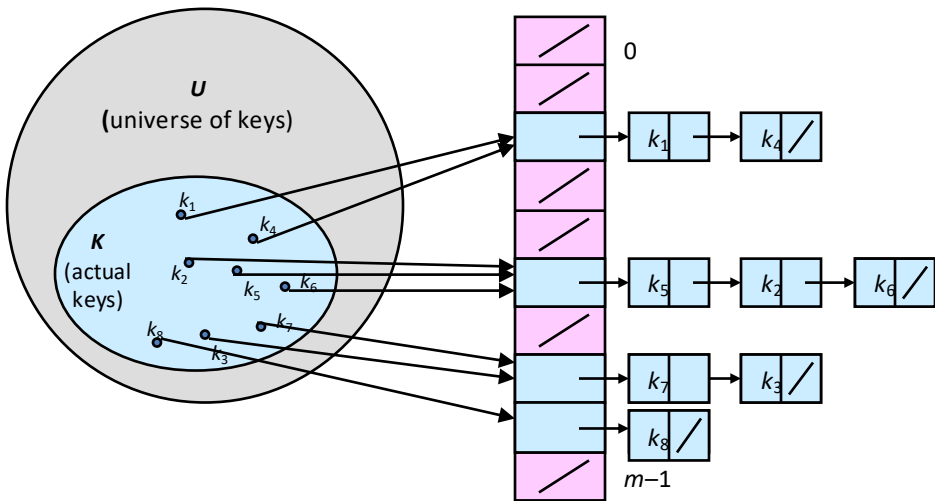
Definition

When $h(o_1) = h(o_2)$ and $o_1 \neq o_2$, this is a collision.

Collision Resolution by Chaining



Collision Resolution by Chaining



Chaining

0
1
2
3
4
5
6
7

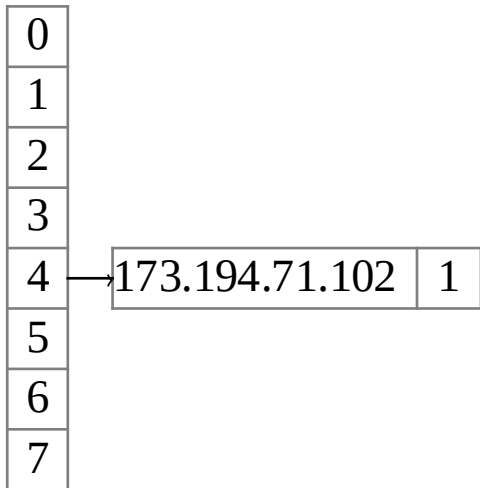
Chaining

$$h(173.194.71.102) = 4$$

0
1
2
3
4
5
6
7

Chaining

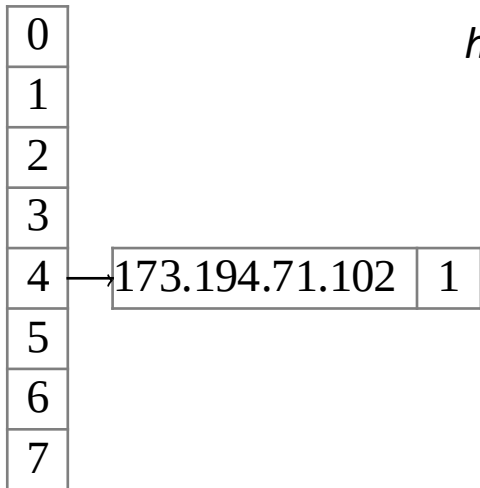
$$h(173.194.71.102) = 4$$



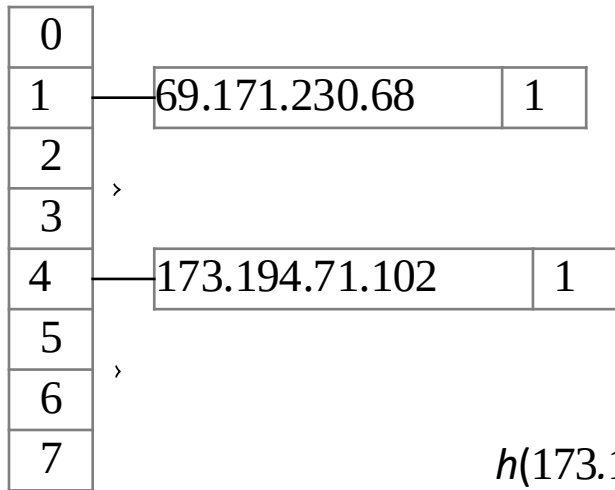
Chaining

$$h(173.194.71.102) = 4$$

$$h(69.171.230.68) = 1$$



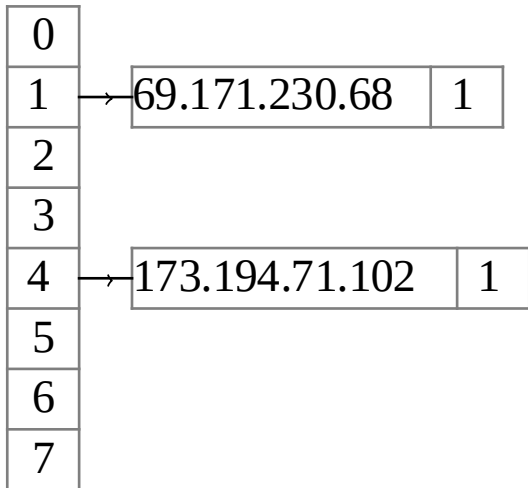
Chaining



$$h(173.194.71.102) = 4$$

$$h(69.171.230.68) = 1$$

Chaining

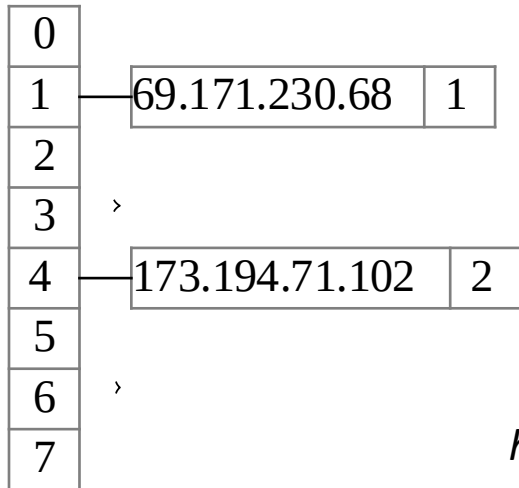


$$h(173.194.71.102) = 4$$

$$h(69.171.230.68) = 1$$

$$h(173.194.71.102) = 4$$

Chaining

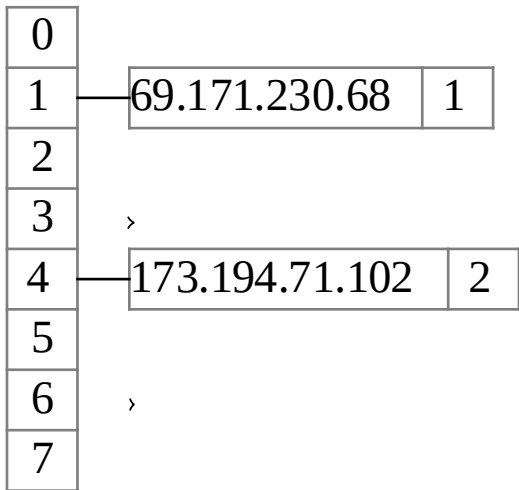


$$h(173.194.71.102) = 4$$

$$h(69.171.230.68) = 1$$

$$h(173.194.71.102) = 4$$

Chaining



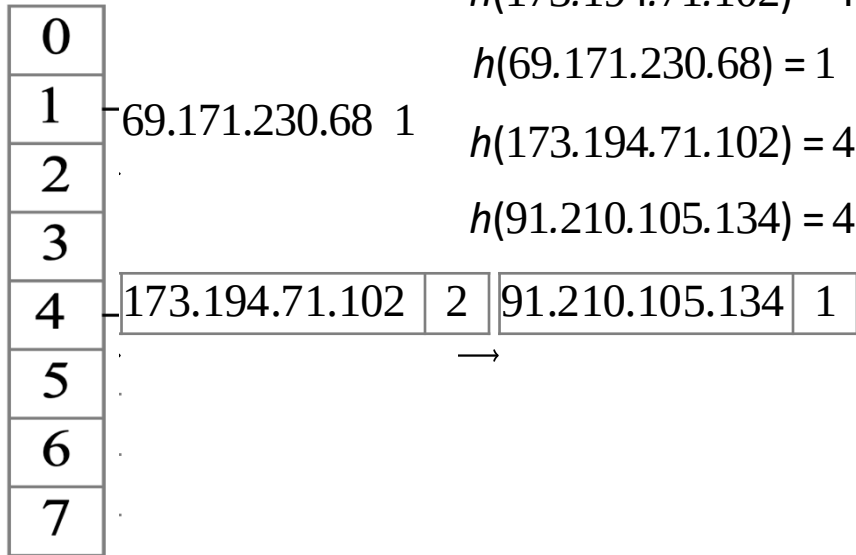
$$h(173.194.71.102) = 4$$

$$h(69.171.230.68) = 1$$

$$h(173.194.71.102) = 4$$

$$h(91.210.105.134) = 4$$

Chaining



Question 1: Using the hash function 'key mod 7', insert the following sequence of keys in the hash table-

50, 700, 76, 85, 92, 73 and 101

Use chaining technique for collision resolution.